Academic year 2025-2026 (Odd Semester)

Date	**Jan 2026**	**Maximum Marks**	**50 + 10**
Course Code	**HS271TA**	**Duration**	**120 Min**
Sem	**VII**	**Improvement CIE**	
INDIAN KNOWLEDGE SYSTEM			

PART – A (10 Marks)

Sl.No		M	CO	BT
1	Which value links IKS with SDGs? A. Consumerism B. Sustainability C. Militarism D. Competition	1	3	1
2	Janapada in Arthaśāstra refers to: A. Army B. Territory and people C. Treasury D. Allies	1	3	1
3	SDGs are adopted by: A. IMF B. World Bank C. United Nations D. WTO	1	3	2
4	The king's happiness lies in: A. Luxury B. Conquest C. Welfare of subjects D. Fame	1	2	1
5	Sustainable governance means: A. Short-term profits B. Resource exploitation C. Intergenerational equity D. Militarization	1	2	1
6	Governance in the Rāmāyaṇa is best described as: A. Autocratic B. Tyrannical C. Welfare-oriented D. Laissez-faire	1	3	1
7	Arthaśāstra views the king as: A. Divine B. Absolute ruler C. Servant of the state D. Merchant	1	2	1
8	The idea of "Lokasaṅgraha" means: A. Individual gain B. Social welfare C. Military power D. Taxation	1	3	1
9	Which ancient text integrates diplomacy, war and peace? A. Upaniṣads B. Rāmāyaṇa C. Arthaśāstra D. Vedas	1	3	1
10	Which is NOT one of the Saptaṅga theory elements? A. Swāmin B. Mantri C. Dharma D. Janapada	1	2	1

TEST (50 marks)

Sl. No.	Part- B Questions	M	CO	BT
1 a	Define Rāja Dharma.	02	4	1
b	What are the key governance attributes highlighted in the Rāmāyaṇa?	02		2
c	Conclude the relevance of ancient Indian governance.	03		2
d	Explain Saptaṅga theory of state.	03		2
2 a	Describe governance principles from Rāma's rule.	03	3	2
b	Compare ancient and modern governance values.	03		2
c	Why is Kosa considered vital for administration?	02		1
d	Name the wives of Bharath and Shathrghan from Ramayana	02		2
3 a	Explain the ideal ruler concept in the Rāmāyaṇa.	03	3	1
b	How does Arthaśāstra emphasize accountability?	03		2
c	What role did ethics play in administration?	02		1
d	What are SDGs?	02		1

Academic year 2025-2026 (Odd Semester)

4 a	Mention the Principles of Effective Governance in the Arthaśāstra	03	4	2
b	Explain the Interrelationship of the Concepts of arthashatra	03		2
c	Mention any two Governance Principles Illustrated in Rāmāyaṇa	02		1
d	Name any two SDGs with the theme	02		1
5 a	What is sustainable governance?	02	4	1
b	What lessons does Rāmāyaṇa offer modern administrators?	03		1
c	How does Dharma relate to SDGs?	03		2
d	What does Durga signify in governance?	02		2

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DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 12th January 2026	**Improvement CIE**	**Max. Marks : 10 + 50**
Semester : 7th	**UG**	**Duration : 30 + 90 Min**
Course Title: Product Design and Development		**Course Code : IM372IA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	Give one example of poor human-machine interaction.	01	L_1	CO2
2.	How does AI support cognitive ergonomics?	01	L_2	CO3
3.	What is workspace optimization using AI?	01	L_2	CO4
4.	What is the main goal of DFM?	01	L_2	CO3
5.	What is meant by the approach to design for manufacturing?	01	L_2	CO4
6.	State one principle of prototyping.	01	L_1	CO4
7.	What is the first step in planning for prototypes?	01	L_2	CO4
8.	What is virtual prototyping?	01	L_2	CO4
9.	What is concurrent engineering?	01	L_1	CO4
10.	Give one advantage of virtual prototyping over physical prototyping.	01	L_2	CO3
	Part – B			
1.	Consider any of the following products and well-known brand related to kitchen supplies, tools, office supply, gifts and a set of competing products. (For Example, a Traditional Cork Screw Bottle opener and a Big Wing Corkscrew Bottle opener). Assess both the brand and competing product on five ID quality categories such as Quality of the user interaction, Emotion Appeal, Ability to Maintain and Repair the Product, Appropriate use of resources and Product Differentiation. Use levels of Low, Medium and High for the assessment to identify the Level of Importance in each of the five categories. Which product would you purchase? Would you be willing to pay more for it than for the others?	10	L_4	CO4
2.	What are the four purposes for which prototyping can be used? Discuss each of these types briefly. A furniture manufacturer was considering a line of seating products to be fabricated by cutting and bending a recycled plastic product available in large sheets. Create a Sketch of the prototype of at least one possible chair design assuming paper or cardboard as prototyping material. What can you learn about the chair design?	10	L_4	CO3

3	What are the benefits of developing the physical prototypes during design and development of procts? Identify and position the following prototypes i. Boeing wing prototype ii. Bahco screwdrivers iii. Projector iv. Bicycle in SolidWorks v. Urethane casting in the plot along the four quadrants along X axis (Ranging from Focussed to Comprehesive) along Y axis (Rannging from physical to analytical)Briefly expl*[illegible]* your reasoning for each placement.	10	L_4	CO3
4.	Explain the differences between a "works-like"ing prototype and a "looks- alike "prototype with a sketch of both for a product like a Nail cutter or any other consumer product of your choice. (Showing the detailing in the sketch is absolutely mandatory) Many product development teams separate the "looks-like" prototype from the "works-like" prototype. They do this because integrating both function and form is difficult in the early phases of development. What are the strengths and weaknesses of this approach? For what types of products might this approach be dangerous?	05	L_4	*[illegible]*
5a.	What are the principles of human-machine interaction, and how they are applied in a medical diagnostic kiosk?	05	L_4	*[illegible]*
b.	How is ergonomics integrated into IoT-based smart kitchen appliances?	05	L_4	*[illegible]*

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Date : 13/01/2026	**Improvement CIE**	**Max. Marks : 10 + 50**
Semester : VII	**UG**	**Duration : 30 + 90 Min**
Course Title: Total Quality Management		**Course Code : IM373TA**

Note: Answer all questions

Sl. No	Questions	M	BT	CO
	Part – A			
1.	Define DPMO.	2	1	CO1
2.	What is the primary goal of Six Sigma?	2	1	CO3
3.	What is the role of a Champion in Six Sigma?	2	2	CO2
4.	List the five phases of the standard Six Sigma improvement methodology.	2	2	CO3
5.	Why is verification important in DFSS?	2	3	CO4
	Part – B			
1	Explain the historical development of Six Sigma.	10	2	CO2
2	Who are the key players in a Six Sigma project and what are their roles? Enumerate.	10	3	CO2
3	Explain how Failure Modes and Effects Analysis (FMEA) supports Design for Six Sigma implementation.	10	2	CO2
4	Choose a product and explain how do you apply Design for Six Sigma in product development?	10	2	CO4
5	Explain how Design for Six Sigma can be applied to service industries.	10	2	CO4

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Date : 12th JAN 2026	**IMPROVEMENT CIE**	**Max. Marks : 10 + 50**
Semester : VII	**UG**	**Duration : 30 + 90 Min**
Course Title: Artificial Intelligence in Industrial and Systems Engineering		**Course Code : IM374TFB**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1	Define AI-based ergonomic risk assessment.	2	3	02
2	What is pose estimation in human motion analysis?	2	2	01
3	Mention two benefits of human–robot collaboration.	2	2	03
4	What is ergonomic risk assessment using AI?	2	3	01
5	Define human–robot collaboration (HRC).	2	2	02
	Part – B			
1	Explain intelligent assistance systems and their role in supporting industrial workers.	10	3	01
2	How does AI help in evaluating task performance in industrial environments?	10	3	02
3	Explain how AI improves safety in human–robot collaboration.	10	3	03
4	Explain how AI-powered human motion analysis improves ergonomic evaluation.	10	4	03
5	Discuss the role of computer vision in ergonomic risk assessment.	10	3	03

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Date : 12th Jan 2026	**CIE – (Improvement)**	**Max. Marks : 10 + 50**
Semester : 7th	**UG**	**Duration : 30 + 90 Min**
Course Title: Predictive Analytics		**Course Code : IM374TFA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	An industrial engineering firm is using sensor data from its machinery to predict time-to-failure (a regression problem). The data contains thousands of highly correlated features and noise. Which ensemble method is generally preferred for achieving high predictive accuracy in complex, noisy, non-linear industrial applications?	02	L_3	CO3
2.	A financial management firm uses an ensemble model to classify loan applications as high-risk or low-risk. Due to strict regulatory requirements, they need to ensure the final model is highly robust and its prediction variance is minimized, even if individual decision trees are slightly biased. Which ensemble method is inherently designed for variance reduction through parallel training on bootstrapped data?	02	L_3	CO3
3.	What is the goal of Principal Component Analysis?	01	L_1	CO3
4.	Anomaly detection aims to identify what in the Data pattern?	01	L_2	CO4
5.	Which data mining technique is primarily used to discover relationships between items in large transaction datasets, such as finding products frequently bought together in a retail setting?	01	L_2	CO3
6.	The first common first step in integrating a data mining process into an existing industrial workflow, after defining the problem, is to assess the quality of what and availability of what with respect to relevance to industrial scenario?	01	L_1	CO4
7.	A company wants to predict which machine in a factory is most likely to fail in the next 7 days based on sensor data. What is the most appropriate *supervised* data mining technique for this **predictive maintenance** problem?	02	L_4	CO4
	Part – B			
1a.	Covariance matrix: $\Sigma = \begin{matrix} 4 & 2 & 0 \\ 2 & 3 & 0 \\ 0 & 0 & 1 \end{matrix}$ Compute eigenvalues and proportion of variance explained by the first two principal components.	10	L_4	CO3
2a.	Throughputs: ([95, 100, 98, 97, 96, 150, 99, 101]). Compute (Q1, Q3), IQR, and flag outliers using (x < Q1 - 1.5* *mathrm{IQR}) or (x > Q3 + 1.5*cdot *mathrm{IQR}).	06	L_3	CO4
b.	A company is analyzing customer feedback from Twitter. Which specific sub-field of **Text Mining** would be most directly applied to automatically classify these tweets as positive, negative, or neutral?	04	L_4	CO3

3	Critically evaluate the statement: "For any supervised learning problem, if a Random Forest and a Gradient Boosting Machine are both meticulously tuned, the Gradient Boosting Machine will theoretically always achieve superior performance." Your evaluation must discuss the inherent model structure and specific data conditions (e.g., noise, outliers) where this statement might fail.	10	L_4	CO4
4a.	In a financial fraud detection case study, a high volume of false positives (labeling legitimate transactions as fraudulent) is observed. Propose one simple strategy to adjust the model or its outputs to reduce the false positive rate.	05	*[illegible]*	*[illegible]*
b.	Assess the primary ethical consideration raised when using credit card transaction data (even if anonymized) in a finance industry case study for building a predictive model of spending habits.	05	*[illegible]*	*[illegible]*
5a.	Design and justify the most appropriate data mining technique (classification, clustering, or association rule mining) for a healthcare insurance company aiming to identify groups of fraudulent claim patterns that they have never seen before.	06	*[illegible]*	*[illegible]*
b.	You are integrating a demand forecasting model (regression) into a supply chain's existing inventory management system. Justify the need for a rigorous A/B testing phase before full deployment of the new model.	04	*[illegible]*	*[illegible]*

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Date : 12th January 2025	**Improvement CIE**	**Max. Marks : 10 + 50**
Semester : 7th	**UG**	**Duration : 30 + 90 Min**
Course Title: Sustainability Engineering & Circular Economy		**Course Code : IM374TFD**

Note:
1. Answer all the Questions.

PART – A

Sl. No.	Questions	M	BT	CO
1	State two economic advantages of Circular Economy.	2	2	4
2	What does the 7R model stand for?	2	2	4
3	What is the significance of "Design for Disassembly"?	2	2	4
4	Define resource efficiency in the context of Circular Economy.	2	1	4
5	What is the link between Circular Economy and SDG 12?	2	2	1

PART – B

Sl. No.	Questions	M	BT	CO
1.	Discuss the integration of Sustainable Engineering with Circular Economy.	10	2	1
2.	Describe the framework and design strategies of Circular Economy.	10	2	4
3.	Propose a Circular Eco-Design for a common product- Smart phone.	10	5	4
4.	Discuss the role of Circular Economy in shaping future environmental governance.	10	3	4
5.	Propose a Circular Economy strategy for garment manufacturing sector.	10	5	4

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DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 14/01/2026	**CIE – Improvement**	**Max. Marks : 10 + 50**
Semester : VII	**UG**	**Duration : 30 + 90 Min**
Course Title: Global Supply Chain Management		**Course Code : IM375TGN**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	A smartphone manufacturer decides to procure chargers from an external vendor instead of producing them internally. This is an example of a _____ decision.	1	2	3
2.	In supplier selection, _____ are used to create competition among suppliers to achieve lower prices.	1	2	3
3.	Revenue management is especially important for _____ assets such as airline seats and hotel rooms.	1	1	1
4.	Vendor-Managed Inventory (VMI) is a replenishment approach where the _____ is responsible for managing inventory at the customer location.	1	2	3
5.	Sharing risk and reward among supply chain partners helps improve _____ and overall performance.	1	1	3
6.	A manufacturer experiences large order fluctuations despite stable customer demand due to poor information sharing. This phenomenon is called the _____ effect.	1	1	1
7.	A retail company hires an external firm to manage transportation, warehousing, and order fulfilment. This firm is best described as a _____ provider.	1	2	3
8.	A supermarket allows its supplier to monitor sales data and replenish stock automatically to avoid stockouts. This system is known as _____.	1	2	3
9.	An airline increases ticket prices as the departure date approaches due to limited seat availability. This practice illustrates _____ pricing.	1	2	1
10.	A buyer and supplier jointly agree to share profits from cost savings achieved through collaboration. This represents _____ sharing in the supply chain.	1	1	3
	Part – B			
1	Compare Vendor-Managed Inventory (VMI) and Collaborative Planning, Forecasting, and Replenishment (CPFR), highlighting their roles in achieving supply chain coordination.	10	3	3
2	Describe the Supply Chain IT framework and explain how the integration of planning, execution, and analytics systems helps organizations achieve competitive advantage.	10	2	4
3	Explain the bullwhip effect in supply chains and critically discuss comprehensive strategies adopted to minimize its impact.	10	3	1
4	Elaborate on the make-or-buy decision in sourcing and develop a decision framework incorporating strategic, financial, and operational considerations.	10	4	3
5	Explain pricing and revenue management strategies in supply chains and discuss their role in balancing demand and supply.	10	2	3

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Academic year 2025-2026 (Odd Sem)

DEPARTMENT OF
CIVIL ENGINEERING

Date	**14.1.2026**	**Maximum Marks**	**60**
Course Code	**CV375TGG**	**Duration**	**90+20 Min**
Sem	**VII**	**Test-3 (Improvement Test)**	
SOLID WASTE MANAGEMENT & STATUATORY RULES			

PART A

Q. No	Question	Marks	CO	BTL
1	Define E-waste.	2	3	1
2.	Minimum stack height for bio-medical waste incinerator should be ____ mtrs above ground.	1	4	1
3	The thickness of carry bags according to 2016 amendment for plastic waste (management & handling) rules, is _____	1	4	2
4	E-waste (management & handling) rules, was promulgated in the year ____	1	3	1
5	Define Extended producer responsibility as per E- waste rules.	1	3	2
6	The temperature of primary chamber in the biomedical waste incinerator shall be _____	1	4	1
7	Schedule ____ of Bio-medical waste management rules corresponds to label for transport of Bio-medical waste containers or bags.	1	4	1
10	Enumerate the equation to calculate the combustion efficiency of the biomedical waste incinerator.	1	4	1
11	What are multi layered plastics.	1	3	1

PART B

Q. No.	Questions	Marks	Course Outcomes	BTL
1	Explain briefly different categories of Bio-medical waste and their treatment/disposal options.	10	4	1
2	Enumerate the salient features of E-waste (management & handling) rules, 2016.	10	3	1
3	a. State the adverse effects of plastic waste. b. Illustrate the standards for deep burial of bio- medical waste.	5 5	3 4	2 1
4	Explain the roles and responsibilities of E-waste producer, recycler and dismantler.	10	4	2
5	Discuss the salient features of plastic waste (management & handling) rules, 2016 with amendments.	10	4	*[illegible]*

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DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 14 Jan 2026	**Improvement CIE**	**Max. Marks : 10 + 50**
Semester : VII	**UG**	**Duration : 30 + 90 Min**
Course Title: Principles & Practices Of Cyber Law		**Course Code : HS375TGT**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	What is meant by "Right to be Forgotten" under GDPR?	2	2	2
2.	Define Data Breach?	2	2	2
3.	What is the main objective of PIPEDA?	2	2	2
4.	List any two challenges to intellectual property (IP) protection in cyberspace?	2	2	4
5.	An E-Commerce site uses a software method to classify products based on customer behavior. Why will the Indian Patent Office treat this as a business method?	2	2	4
	Part – B			
1	Analyse the rights granted to data subjects under the GDPR. How do these rights strengthen informational privacy and digital autonomy?	10	3	2
2	A hospital uses a cloud-based system to store patient records, including biometrics, medical history, Aadhar details, and insurance information. A misconfigured server allowed public access for several days before discovery. Explain the types of data exposed in this incident. Analyze why protecting such data is critical in cyberspace and discuss the legal liabilities under the IT Act and the Digital Personal Data Protection Act, 2023.	10	3	2
3	A website hosts thousands of pirated e-books and music files without permission from copyright owners. The domain name closely resembles a famous publisher's trademark, and the site also uses a customized algorithm to hide copyright-protected content. Discuss the IP issues involved and identify legal remedies available under Indian law.	10	4	4
4	A mobile app mimics the interface and services of a competing e-commerce platform. It copies slogans, product images, and uses a domain name with minor spelling variation. Users are tricked into believing it is the official app. Analyse the copyright, trademark, and unfair competition issues. Suggest legal remedies.	10	4	4
5	Discuss the major trademark issues that arise in cyberspace. Explain the distinction between domain names and trademarks, the nature of domain name disputes and the relevant legal framework governing them.	10	3	4

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DEPARTMENT OF MECHANICAL ENGINEERING AY 2025-26 (ODD SEM)				
Date	**13/01/2026 Tuesday 1.20 PM to 3.20 PM**	**Improvement CIE**	**Max. Marks**	**10+50**
Semester	**VII**	**UG**	**Duration**	**110 mins**
Course	**INDUSTRY 5.0 (Institutional Elective)**		**Code**	**ME375TGP**

#	QUIZ	M	BT	CO
1	What role do cobots (collaborative robots) play in Industry 5.0?	1	1	1
2	How does Industry 5.0 promote sustainability?	1	1	1
3	Which technology enables personalized manufacturing in Industry 5.0?	1	2	2
4	A potential challenge associated with adopting Industry 5.0 is ————-	1	2	2
5	How does Industry 5.0 redefine the role of humans in manufacturing compared to Industry 4.0?	1	2	2
6	In the context of Industry 5.0, what does “mass personalization” refer to?	1	1	1
7	A critical step in the Industry 5.0 transition roadmap is workforce development. What does this primarily involve?	1	2	2
8	Why is energy-efficient and “green” AI a key research direction for Industry 5.0?	1	1	1
9	What is the purpose of a “digital twin” in the context of Industry 5.0 research and trends?	1	2	2
10	In the aerospace sector, Boeing has successfully trialed which Industry 5.0-aligned technology to reduce worker fatigue and increase productivity?	1	2	2

#	TEST	M	BT	CO
1	Bring out the important differences between the Industry 4.0 & Industry 5.0 frameworks. Also mention the benefits of the transition from Industry 4.0 to 5.0.	10	3	1
2	Elaborate the key characteristics, technologies & benefits of Industry 4.0.	10	4	2
3	Name the three core pillars of Industry 5.0 and explain the key technologies actions involved.	10	2	2
4	Illustrate some case studies of successful implementation of Industry 5.0 from automotive, aerospace, and electronics industries.	10	3	2
5	Summarize the research directions that are ongoing to integrate industrial innovation with broader societal values.	10	4	3

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Academic year 2025-2026(ODD Sem)

DEPARTMENT OF
ELECTRONICS AND INSTRUMENTATION ENGINEERING

Date	**14.01.2026**	**Maximum Marks**	**10+50**
Course Code	**EI375TGK**	**Duration**	**120 Min**
Sem	**VII**	**Improvement CIE**	
Course Title: DISEASE AND DIAGNOSTICS AN ENGINEERING PERSPECTIVE			

Sl. No.	PART-A	M	BT	CO
1.1	What is Artificial Intelligence (AI) in healthcare?	2	L1	4
1.2	Give one example of structured data in healthcare.	1	L2	4
1.3	Give one example of unstructured healthcare data.	1	L2	4
1.4	What is data bias in AI?	1	L1	4
1.5	What are the two main components of the CNS?	1	L1	3
1.6	Which test is commonly used to diagnose epilepsy?	1	L1	3
1.7	What type of information does EEG provide that CT and MRI cannot?	1	L2	3
1.8	What is the purpose of the 10–20 system in EEG?	2	L2	3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	-	-	05	05	05	05	-	-	-	-

PART-B

Sl. No.		M	BT	CO
1.	Explain the basic functioning of the Central Nervous System (CNS). Describe its main components and their roles.	10	L1	3
2.	Explain how CT, MRI, and EEG complement each other in CNS diagnosis.	10	L2	3
3.a	List the common ailments of the CNS. Explain any four with their symptoms in brief.	5	L1	3
3.b	Discuss the major challenges in deploying AI in healthcare, focusing on data bias, privacy, and integration into clinical workflow.	5	L2	4
4.	Explain the concept of Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL) in the context of healthcare	10	L1	4
5.	Explain the types of data used in healthcare. Differentiate between structured and unstructured data with examples.	10	L2	4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	-	-	25	25	25	25	-	-	-	-

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Program	CHEMICAL ENGINEERING			
Course Title	GREEN AND HYDROGEN TECHNOLOGY			
Course Code	CH375TGD			
CONTINUOUS INTERNAL EVALUATION 3 (Improvement Test)				
INSTITUTIONAL ELECTIVE: GROUP D				
Sem	VII	**Date | Time**	**Jan 2026**	
Max Marks	**60**	**14.01.2026 | 13:20 - 15:20**	**Max Duration**	**120 Min**

Instructions to Candidates:

All the questions are compulsory. No additional booklet will be provided.

Q. No.	QUESTIONS	M	BT	CO
1	What is hydrodesulfurization?	02	01	03
2	Define hydrogenation?	02	01	02
3	List the applications of hydrogen in fertilizer industry.	02	01	01
4	What is the use of hydrogen in aviation sector?	02	01	04
5	What is the role of hydrogen in fuel cells?	02	01	04
PART B				
1	How are hydrogen hazards classified? Explain deflagration, detonation, explosion, fireball, and flash fire.	10	02	01
2	List and explain the hydrogen risk components related to material properties and hydrogen handling aspects.	10	01	04
3	List the safety features to be incorporated in electrolyzers, pressure vessels, trailers (gaseous and liquid), pipelines, and fuel cell vehicles for hydrogen handling.	10	03	03
4	Give an explanatory note on the applications of hydrogen in refineries and petrochemicals.	10	02	02
5	Explain the role of hydrogen in transport and automotive sectors.	10	02	02

Course Outcomes

At the end of the course the student will be able to,	
1	Understand the importance of hydrogen and its use as an energy carrier
2	Explain the production, storage and handling of hydrogen
3	Analyze the need for hydrogen as an alternate fuel and the associated challenges
4	Appraise the importance of safety, regulations and codes

Cos/BTL	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
Marks	10	20	10	10	10	30	10	-	-	-

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Department of
Aerospace Engineering

Date	14th January 2026	Maximum Marks	60
Course Code	AS375TGB	Duration	120Min
Sem	VII Semester	Offline: Quiz &Test (Improvement)	
AIRCRAFT SYSTEMS			

PART -A

Sl No	Questions	M	BT	CO
1	List any two fire suppression agents used in aircraft.	2	2	2
2	Why is cabin pressurization important at high altitudes?	2	2	3
3	State one advantage of fully powered flight controls over conventional systems.	2	3	1
4	What is the purpose of trim tabs in flight control systems?	2	4	1
5	Differentiate between primary and secondary flight controls.	2	2	4

PART-B

Sl. No.	Questions	M	BT	CO
1.	Evaluate the design and operation of aircraft fire detection and suppression systems with respect to safety requirements.	10	L4	4
2.	Compare air cycle and vapour cycle air-conditioning systems used in aircraft. Analyze their suitability for different aircraft categories.	10	L3	3
3.	Explain the history of airplanes and its influence on the development of modern flight control systems	10	L2	1
4.	Analyze how redundancy and fault tolerance are incorporated in electronic flight control systems.	10	L3	2
5.	Analyze the limitations of conventional flight control systems and justify the need for power-assisted controls.	10	L3	4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	14	12	12	22	00	14	34	12	0	

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Date: 14-01-2026	CIE 3	Max. Marks: 10+50
Semester: VII	UG	Duration: 2 Hrs (1:20PM- 3:20PM)
Course Title: Mathematics of Music		Course code: MA375TGR

Department of Mathematics

S No	Quiz	M	BT	CO
1.	Define avartana and sama in the context of the Carnatic tala system.	2	1	2
2.	State the tihai formula 3P + 2G = T and explain what each variable represents.	2	2	1
3.	What is a fractal? Give one example of a deterministic fractal.	2	2	2
4.	Define critical band in psychoacoustics and state its significance for consonance perception.	2	3	3
5.	Calculate the total number of aksharas in Misra-jati Rupaka Tala.	2	2	3

S No	Test	M	BT	CO
1a	Explain the hierarchical organization of time in Carnatic music: matra, akshara, and avartana. How do these relate mathematically through jati and nadai?	5	2	1
1b	A tala has structure I O I I with Khanda jati. Calculate the number of aksharas in one cycle. If rendered in Misra nadai, how many matras are in one avartana?	5	2	1
2a	Design a tihai for 2 cycles of Adi Tala in Chatusra nadai (64 matras). Specify the pattern length P and gap G using the tihai formula 3P + 2G = T. Verify that your design ends at sama. (ii)	5	3.	2
2b	A korvai has purvanga of 12 matras and uttaranga of 8 matras. If repeated 3 times, calculate the total length. Does it fit exactly in 2 cycles of Rupaka Tala (Chatusra jati, Chatusra nadai)? Justify.	5	3	2
3a	Explain the concept of 1/f noise (pink noise) in music as discovered by Voss and Clarke. How does it relate to entropy and redundancy in musical composition?	5	3	4
3b	Calculate the Shannon entropy and redundancy for a four-note melody system with probabilities: Sa (0.4), Ri (0.3), Ga (0.2), Ma (0.1). Interpret your results in terms of listener engagement.	5	3	4
4	(i) Define critical bandwidth using Zwicker's formula. Calculate the critical bandwidth at 520 Hz. (ii) Using your result from (i), analyze whether an equal temperament minor third interval (520 Hz and _____ Hz) would be perceived as consonant or dissonant. Explain the relationship between critical bands and the perception of roughness.	*[illegible]*	*[illegible]*	*[illegible]*
5a	Define a first-order Markov chain. Construct a valid 3×3 transition matrix for generating melodies using notes {C, D, E} where stepwise motion has probability 0.5, staying on the same note has probability 0.3, and skipping has probability 0.2. Verify your matrix is row-stochastic.	*[illegible]*	*[illegible]*	*[illegible]*
5b	Explain the missing fundamental phenomenon in psychoacoustics. A complex tone contains harmonics at 200 Hz, 300 Hz, 400 Hz, and 500 Hz but lacks the fundamental. What pitch will a listener perceive and why?	*[illegible]*	*[illegible]*	*[illegible]*

[The M/BT/CO columns of Q4 and Q5 are cut off in the scanned copy.]

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Department of Electronics and Telecommunication Engineering

Date: 14/01/2026	**CIE - 3**	**Max. Marks: 60**
Semester: 7	**UG**	**Duration: 2 Hrs**
Course Title: Space Technology and Applications		**Course Code:ET375TGL**

Q.No	QUIZ	M	BT	CO
1	List the different types of remote sensing systems based on the range of electromagnetic radiation.	2	L2	CO3
2	Compare the active and passive remote sensing systems	2	L3	CO3
3	What are the main functions of the space shuttle? List its main components.	2	L2	CO4
4	What are the design constraints of the planetary payloads?	2	L2	CO4
5	What are the communication challenges faced during lunar missions?	2	L2	CO4
	TEST			
1	Explain the working principle of passive remote sensing systems along with advantages and limitations.	10	L2	CO3
2	Explain the following with respect to satellite image i) Spatial Resolution ii) Spectral Resolution iii) Temporal Resolution iv) Radiometric Resolution	10	L2	CO3
3	Explain the Image processing techniques to process the satellite data into useful information.	10	L2	CO3
4	Illustrate the following applications of remote sensing satellites i) Land Cover Classification ii) Land Cover Change Detection	10	L3	CO3
5	Explain the Deep space planetary mission along with challenges and requirements.	10	L2	CO4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	-	-	44	16	-	48	12	-	-	-

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DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**14.01.2026**	**Maximum Marks**	**10 + 50**
Course Code	**IM375TGM**	**Duration**	**120 Mins**
Sem	**VII Semester**	**Improvement CIE**	
PROJECT MANAGEMENT (Institutional Elective)			

Note:
1. Answer all the Questions.

PART – A

Sl. No.	Questions	M	BT	CO
1	Name any two project scheduling techniques.	02	L1	CO1
2	Differentiate between CPM and PERT	02	L2	CO1
3	List any two positive strategies to mitigate risk.	02	L2	CO2
4	Highlight any two quantitative risk analysis tools.	02	L1	CO1
5	Mention any two tools used in qualitative risk analysis.	02	L2	CO2

PART – B

Sl. No.	Questions	M	BT	CO
1.	Draw the network diagram and find forward pass & backword pass computations, identify critical path for the following: Activity: 1 – 2 | 1 – 3 | 1 – 4 | 2 – 5 | 3 – 7 | 4 – 6 | 5 – 7 Duration: 10 | 8 | 9 | 8 | 16 | 7 | 7 Activity: 5 – 8 | 6 – 7 | 6 – 9 | 7 – 10 | 8 – 10 | 9 – 10 Duration: 6 | 7 | 5 | 12 | 13 | 15	15	L4	CO4
2.	A manufacturing company is launching a new product and needs to plan its project schedule efficiently. As a project manager, explain the steps involved in project implementation scheduling and discuss how effective time management and resource allocation methods can optimize the project timeline.	10	L2	CO2
3.	The following network diagram represents activities associated with a project: Activity: A | B | C | D | E | F | G | H | I Optimistic Time: 5 | 18 | 26 | 16 | 15 | 6 | 7 | 7 | 3 Pessimistic Time: 10 | 22 | 40 | 20 | 25 | 12 | 12 | 9 | 5 Most Likely Time: 8 | 20 | 33 | 18 | 20 | 9 | 10 | 8 | 4 i. Find expected completion time and variance of each activity ii. Draw Network diagram iii. The critical path and project duration iv. The duration of the project that will have 95 per cent chance of being completed.	15	L4	CO4
4.	Explain the application of Project Life Cycle Costing in an enterprise software implementation project.	10	L4	CO4

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DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**17th January 2025**	**Maximum Marks**	**10 + 50**
Course Code	**21HS71**	**Duration**	**120 Mins**
Sem	**VII Semester**	**Improvement CIE**	
CONSTITUTION OF INDIA AND PROFESSIONAL ETHICS			

Note:
1. Answer all the Questions.

PART – A

Sl. No.	Questions	M	BT	CO
1	_____ is termination by the employer of the service of a worker for any reason whatsoever, otherwise than as a punishment inflicted by way of disciplinary action.	01	2	2
2	The _____ Act provides certain benefits to employees in case of sickness, maternity and employment injury, related matters.	01	1	2
3	The working hours of an adolescent worker shall be regulated in accordance with the provisions of the _____ Act.	01	2	2
4	List any two Roles Played By the CSR Committee	01	2	2
5	List any two objectives of Industrial policy in India	02	2	2
6	What is child labor?	02	1	3
7	What is Fault Tree Analysis Approach?	02	3	1

PART – B

Sl. No.	Questions	M	BT	CO
1.	Define Corporate Social Responsibility? Explain the role of and responsibilities of CSR in society with two cases.	10	2	4
2.	Discuss the Guidelines stipulated for Universities and colleges to curb the menace of ragging.	10	1	3
3	Explain the prohibition and prevention of sexual harassment of women employee in working place with decided case study.	10	3	3
4	Highlight the thrust areas of Government in framing labour laws and list various labour legislation laws enacted by Central Government.	10	2	2
5	Define Industrial dispute. Explain the mechanism for resolving Industrial disputes under the Industrial dispute Act 1947.	10	2	4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Max Marks	Quiz	–		–	–	–	–	10	–	–	–	–
		Test	–	10	20	20	–	10	30	10	-	-	–

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Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**17th February 2025**	**Maximum Marks**	**10 + 50**
Course Code	**21IM72**	**Duration**	**120 Mins**
Sem	**VII Semester**	**Improvement CIE**	
PRODUCT DESIGN & DEVELOPMENT			

Note:
1. Answer all the Questions.
2. Avoid verbose answers and answer specifically and to the point.
3. Sketches must be provided in Part B answers as appropriate.
4. Engineering specifications and Qualitative and/or Quantitative metrics must be mentioned in Part B answers appropriately.

PART – A

Sl. No.	Questions	M	BT	CO
1	What is the role of Monte Carlo simulation in risk analysis for product development projects?	01	L_2	CO4
2	How does Design Structure Matrix (DSM) help in managing dependencies in iterative product design projects?	01	L_2	CO4
3	Name one key risk associated with product design and development projects.	01	L_2	CO4
4	What is meant by “design parallelism” in Concurrent Engineering?	01	L_2	CO3
5	What impact does Concurrent Engineering have on reducing design iterations and rework?	01	L_2	CO3
6	How does Industrial Design differ from Engineering Design?	01	L_2	CO3
7	Why is Biomechanics important in Ergonomics for product design?	01	L_2	CO3
8	How do designers use eye-tracking technology in Ergonomic studies?	01	L_2	CO4
9	How does the Fitts’ Law apply to Ergonomic design in interactive products?	01	L_2	CO4
10	What is the importance of Grip Design in handheld product development?	01	L_2	CO4

PART – B

Sl. No.	Questions	M	BT	CO
1.	How does cross-functional team collaboration improve the efficiency of product development projects? Provide real world industry based examples.	10	L_4	CO3
2.	Explain how simultaneous product and process development helps improve efficiency in Concurrent Engineering. Discuss with two case examples.	10	L_3	CO3
3.	Explain how 3D Printing and Additive Manufacturing are transforming Industrial Design. Discuss advantages and limitations, Discuss using examples.	10	L_3	CO4
4.	Discuss how sports equipment design (e.g., running shoes, bicycles) integrates Industrial Design and Ergonomics for better performance. Specific product features covering performance, ergonomics and industrial design must be illustrated with appropriate sketches.	10	L_4	CO4
5.	Explain the role of sustainability in Industrial Design. How can sustainable design principles be integrated into product development? Give two case examples with supporting illustrative sketches.	10	L_4	CO4

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BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

	Particulars	CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
Marks Distribution		*[illegible]* (cut off in the scan)										

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DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**19th February 2025**	**Maximum Marks**	**05 + 25**
Course Code	**21IM78**	**Duration**	**60 Mins**
Sem	**VII Semester**	**Improvement CIE**	
INNOVATION AND ENTREPRENEURSHIP			

Note:
1. Answer all the Questions.
2. Use suitable sketches and tabular column wherever necessary.
3. Observe the marks split-up and answer accordingly.

PART – A

Sl. No.	Questions	M	BT	CO
1	Comment on the importance of Maximization with regard to pricing strategies.	01	L1	CO1
2	Provide two examples business that you can start now by using your existing skills and generate revenue with bootstrapping.	01	L4	CO2
3	Identify any two secondary revenue streams.	02	L1	CO2
4	Comment on how startups have used "Content Marketing" effectively to bring awareness to their offerings with an example.	01	L3	CO3

PART – B

Sl. No.	Questions	M	BT	CO
1.	Write short notes on i. Segmentation and Targeting ii. Importance of Early adopters	5	2	2
2.	Suggest a solution demo for any two of the following companies: i. A company building houses on wheels ii. A company manufacturing grocery bags that dissolve in water iii. A company that has made an app for visually challenged people iv. A company that rents furniture	2.5 *2 = 5	3	3
3.	Explain how Minimum Viable Product helps in Potential Risk reduction.	5	3	4
4.	Taking an example of Cirque De Sole or any other example of your choice, Illustrate the 4-action framework by plotting strategy canvas to take advantage of uncontested market.	5	3	4
5.	What are the key benefits of using social media marketing for brand engagement?	5	2	4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Max Marks	Quiz	01	03	01	–	–	01	01	01	02	–	–
		Test	–	05	12	08	–	–	05	05	05	10	–

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Date	**18th February 2025**	**Maximum Marks**	**10 + 50**
Course Code	**21IM73G1**	**Duration**	**120 Mins**
Sem	**VII Semester**	**Improvement CIE**	
DATA MINING AND PREDICTIVE ANALYTICS (Elective)			

Note:
1. Answer all the Questions.
2. Avoid verbose answers and answer specifically and to the point.

PART – A

Sl. No.	Questions	M	BT	CO
1	Why are distance metrics important in clustering algorithms?	01	L_2	CO1
2	What is a dendrogram, and how is it used in hierarchical clustering?	01	L_2	CO1
3	What is the significance of "random" in Random Forest?	01	L_2	CO1
4	Can Random Forest be used for both classification and regression tasks?	01	L_2	CO1
5	How does Gradient Boosting differ from Random Forest?	01	L_2	CO1
6	What is the role of the learning rate in Gradient Boosting?	01	L_2	CO2
7	How does anomaly detection differ from outlier detection?	01	L_3	CO2
8	Describe one advantage of using unsupervised learning for anomaly detection.	01	L_3	CO2
9	Name one advantage of using ensemble methods over a single predictive model.	01	L_1	CO2
10	What does the term "bagging" refer to in ensemble techniques	01	L_1	CO2

PART – B

Sl. No.	Questions	M	BT	CO
1.	Consider five data points in 1D: A=1 A = 1, B=3 B = 3, C=4 C = 4, D=8 D = 8, and E=10 E = 10. i. Compute the pairwise Euclidean distances between these points. ii. Using single linkage, perform agglomerative clustering until you have two clusters. List the clusters at each step. *Hint:* In single linkage, the distance between two clusters is the minimum distance between any pair of points in the two clusters.	10	L_2	CO1
2.	You are building a decision tree for a binary classification problem where you have the following data at a node: Class: 0, Count: 30 Class: 1, Count: 70 i. Calculate the Gini impurity for the current node. ii. Suppose the dataset is split into two groups, one with 10 class 0 and 30 class 1, and the other with 20 class 0 and 40 class 1. Calculate the Gini impurity for each of the two splits and compute the weighted average Gini impurity for the entire split.	10	L_2	CO2

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3.	Suppose you have a binary classification problem with the following dataset: x1 | x2 | y 2 | 3 | 0 1 | 4 | 1 3 | 2 | 0 4 | 3 | 1 You decide to use Gradient Boosting Classifier with decision stumps as base learners. For simplicity, let's use a learning rate of η=0.1 and fit one shallow tree (stump) at each boosting iteration.	05	L_3	CO2
4.	Describe the concept of Text Mining and how it is applied in extracting meaningful information from large volumes of unstructured text data. Provide an example of a text mining application.	15	*[illegible]*	*[illegible]*
5.	You are given a dataset containing the transaction details of a credit card company. The dataset includes the following features: transaction amount, transaction time, transaction location, and the customer's previous transaction history. Define what anomaly detection is and explain its importance in the context of credit card fraud detection. Give one more example based on the above description.	10	*[illegible]*	*[illegible]*

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Particulars			CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
Marks Distribution	Max Marks	Quiz	05	05	–	–	–	02	06	02	–	*[illegible]*	*[illegible]*
		Test	10	40	–	–	–	–	35	05	10	*[illegible]*	*[illegible]*

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Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF

INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**18th February 2025**	**Maximum Marks**	**10 + 50**
Course Code	**21IM73GC**	**Duration**	**120 Mins**
Sem	**VII**	**Improvement CIE**	
VIRTUAL COMMERCE (Department Elective)			

Note: Answer all the Questions.

PART – A

Sl. No.	Questions	M	BT	CO
1	What are the key limitations of the current internet infrastructure in meeting the increasing demands of modern society?	02	L2	CO1
2	How can decline costs of IoT devices be applied to maintain reliability and minimize error rates in a large-scale deployment?	02	L2	CO2
3	What is the role of internet access drones in improving internet connectivity, particularly in remote areas?	02	L1	CO3
4	What is the impact of fiber optics in the First Mile on the overall performance of internet services, considering bandwidth and scalability.	02	L2	CO2
5	Compare the strengths and weaknesses of mobile internet access vs. telephone-based internet in the Last Mile, focusing on reliability.	02	L2	CO3

PART – B

Sl. No.	Questions	M	BT	CO
1.	Analyze the limitations of the current internet infrastructure in handling the increasing demands of modern society. Evaluate how the introduction of Internet2® could address these limitations. Furthermore, assess the impact of government regulation and surveillance on the internet, considering both societal benefits and privacy concerns. How do these regulations influence the development of future internet technologies, and what balance should be struck between security and privacy?	10	L4	CO4
2.	Analyze the challenges and advantages of deploying fiber optic technology in the First Mile for handling the bandwidth explosion. How does it impact the scalability and reliability of internet infrastructure? In the context of the Last Mile, compare and evaluate mobile internet access versus telephone-based internet access in terms of speed, accessibility, and reliability. How do these two access technologies influence the overall performance and equity of global internet connectivity?	10	L4	CO4
3.	imagine you are tasked with improving internet latency for a global online gaming platform. Apply your knowledge of future internet latency solutions (such as edge computing, 5G, or satellite networks) to propose a set of strategies that could reduce latency and improve the user experience. How would you prioritize and implement these strategies in a high-traffic gaming environment?	10	L3	CO3

4.	Given the increasing deployment of IoT devices in smart homes and industrial settings, apply your knowledge of guaranteed service levels and lower error rates to design an IoT-based solution that ensures optimal performance and reliability. How would you implement a system that balances low error rates with maintaining the cost-effectiveness of the solution, considering the declining costs of IoT devices?	10	L3	CO1
5.	How has Social Commerce evolved over time? What are its advantages and disadvantages?	10	L2	CO2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Particulars			CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
Marks Distribution	Max Marks	Quiz	05	05	–	–	–	04	06	–	–	–	–
		Test	10	10	10	20	–	10	20	10	10	–	–

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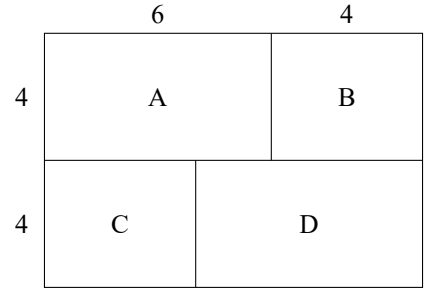
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**18th February 2025**	**Maximum Marks**	**10+50**
Course Code	**21IM74HA**	**Duration**	**120 Min**
Sem	**VII**	**Improvement CIE**	
FACILITIES PLANNING AND DESIGN			

Note:

1. Answer all the Questions.

Q.No	Part – A Questions	M	BT	CO
1.1	List any two basic information required for ALDEP algorithm.	(02)	L1	CO1
1.2	The mission of a warehouse is to effectively _____ products in any configuration to the next step in the supply chain without _____ or altering the product's basic form.	(02)	L2	CO1
1.3	List any two Receiving Principles.	(02)	L1	CO1
1.4	State any two limitations of CRAFT?	(02)	L1	CO1
1.5	Activity relationship provides the basis for ____ in the facilities planning ____.	(02)	L1	CO1

Sl. No.	Part –B Questions	M	BT	CO
1	Consider the following initial layout with unit cost matrix. Use the CRAFT pair wise interchange to obtain the desirable layout.  Initial Layout A B C D A - 1 1 3 B 2 - 2 1 C 0 2 - 2 D 1 4 3 - Flow Matrix	10	L3	CO3
2	Illustrate and provide a summary of the typical warehouse functions and flows.	10	L3	CO2
3	State and explain the steps involved in Automated Layout Design Program (ALDEP).	10	L2	CO3
4	Explain the steps in designing of layout corresponding to typical U Flow for a manufacturing facility.	10	L3	CO4
5	With suitable assumptions propose a layout plan for setting up facilities for an information technology organization.	10	L4	CO4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Particulars			CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
Marks Distribution	Quiz	Max Marks	10	-	-	-	08	02	-	-	-	-
	Test		10	20	20	-	10	30	10	-	-	-

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Date	**19th February 2025**	**Maximum Marks**	**10+50**
Course Code	**21IM74HD**	**Duration**	**120**
Sem	**VII**	**Improvement CIE**	
DATABASE MANAGEMENT SYSTEM			

Note: Answer all the Questions.

Sl No	Questions	M	BT	CO
	Part A			
1.	Identify the basic data types available for attributes in SQL.	(02)	L2	CO1
2.	Mention the guidelines for physical database design.	(02)	L2	CO3
3.	Distinguish between the base relation and virtual relation in SQL environment.	(02)	L3	CO4
4.	"Insertion, deletion and modification anomalies are considered bad in database design". Justify.	(02)	L5	CO1
5.	Mention the expanded form of the acronym BCNF.	(02)	L1	CO3
	Part B			
1	Why is Normalization necessary while designing the relational data base? Analyze various levels in Normalization process.	(10)	L4	CO1
2	Write SQL commands to perform the following: i) To create a table student with fields: USN, SName, Address, Course, Section, TotalMarks ii) To insert values in the above table interactively iii) To create view with fields USN & Name. iv) To change the total marks of 50 if the marks lie in between 47 & 49. v) To delete tuples from the relation if the total marks is less than 50.	(10)	L3	CO1
3	Discuss insertion, deletion, and modification anomalies. Why are they considered bad? Illustrate with examples.	(10)	L4	CO3
4	Consider the following relations for an order-processing application database at ABC. Inc. ORDER (O#, Odate, Cust#, Total_amount) ORDER_ITEM(O#, I#, Qty_ordered, Total_price, Discount%) Assume that each item has a different discount. The Total price refers to one item. Odate is the date on which the order was placed, and the Total_amount is the amount of the order. If we apply a natural join on the relations ORDER_ITEM and ORDER in this database, what does the resulting relation schema look like? What will be its key? Show the FDs in this resulting relation. Is it in 2NF? Is it in 3NF? Why or why not?	(10)	L3	CO4
5 a	Give an overview of different types of JOINS available in SQL.	(06)	L3	CO3
b	Distinguish between JOIN and CARTESIAN PRODUCT.	(04)	L4	CO3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Quiz	Max Marks	04	-	4	2	-	02	04	02	-	02	-
	Test		20	-	20	10	-	-	-	26	24	-	-

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Academic year 2024-2025 (Odd Sem)

Common to all branches

DEPARTMENT OF
BIOTECHNOLOGY

Date	**February, 2025**	**Maximum Marks**	**10 + 50**
Course Code	**21BT75IB**	**Duration**	**120 min**
Sem	**VII Semester**	**Closed Book Offline Quiz and Test – 2**	
Healthcare Analytics (Institutional Elective)			

Answer all questions in Part A & Part B, Answer Part A in first two pages in sequence only

Part A

Sl No	Questions	M	BT-L	CO
1	Write any 2 applications of systems biology.	2	2	1
2	Mass Spectrometry measures _____ charge ratio (m/z) of molecules	1	2	1
3	Mention any 2 steps involved in implementation of systems biology.	2	2	2
4	List any 2 applications of secondary structure prediction.	2	3	2
5	What are the 2 gene prediction methods?	2	2	3
6	Name any tool used to detect the functional sites in a gene	1	4	3

Part B

Sl. No	Questions (Test)	M	BT-L	CO
1.	Elucidate on the gene prediction methods	10	2	3
2	How does the RNA secondary be predicted?	10	3	3
3	Explain the basic structure of a protein.	10	4	4
4	Illustrate on the protein structure predictions.	10	4	4
5	Give a comprehensive note on the protein identity based on the composition.	10	2	3

BTL-Blooms taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks			60	40		40	20	40		

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Academic year 2024-2025 (Odd Sem)

DEPARTMENT OF
CHEMICAL ENGINEERING

Date	**Feb 2025**	**Maximum Marks**	**60**
Course Code	**21CH751C**	**Duration**	**120 Min**
Sem	**VII Semester**	**CIE 3**	
SUSTAINABILITY AND LIFE CYCLE ANALYSIS (Institutional Elective)			

Note: 1. Answer all the Questions.

	Part A	M	BT	CO
[illegible] a	Define the terms odour character and odour persistence.	2	1	2
b	How do you control the odour arising from hydrogen sulphide?	2	1	2
c	Name two fermentative bacteria	2	1	3
d	Write a relation between intensity and concentration,	2	1	3
e	Name two bacteria in acetogenesis.	2	1	2
	Part B			
[illegible]	Explain the working of atomisers to control the odour	10	2	3
[illegible]	State and Explain the stages of biomethanation	10	2	4
[illegible]	With a neat sketch explain the working of fixed film digestion system	10	2	3
[illegible]	What are the stages of bioethanol production? Explain in brief	10	2	3
5	With a neat diagram explain the working of Down draft Gasifier	10	2	4

[The question numbers of Part B Q1–Q4 (left edge) are hidden by the binder ring in the scan.]

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Academic year 2024-2025 (Odd Sem)

(IMPROVEMENT TEST FOR VII SEM)

DEPARTMENT OF CHEMISTRY

Date	19.2.2025	Sem - VII	Improvement test
Course Code	21CM75ID	Maximum Test Marks	Quiz-10 + Test-50 (60M)
Course Name	ACSM	Duration	120 mins
ADVANCES IN CORROSION SCIENCE AND MANAGEMENT			

Note: 1) Answer all the questions

2) Part A should be answered in the first two pages of the answer script.

	Part-A	**M**	**BTL**	**CO**
1	Name the type of corrosion that occurs commonly in oil and gas industries.	1	1	2
2	Arrange the following metals and alloys according to their position in the galvanic series: Gold, Aluminum, Copper, and Magnesium alloy, Bronze Gold, Aluminum, Copper and Magnesium alloy, Bronze.	1	2	2
3	Suggest any one method to reduce seasonal cracking.	1	1	1
4	What is the consequence of hydrogen embrittlement?	1	2	3
5	Pitting is localized and has intense corrosion. Justify.	1	3	2
6	State any one favourable condition required for rusting to take place.	1	2	1
7	List any one disadvantage of a corrosion map.	1	1	2
8	What is the significance of the Tafel equation in corrosion studies?	1	3	2
9	How does electrical isolation help prevent differential metallic corrosion?	1	2	2
10	List any one preventive measure to avoid microbial corrosion.	1	1	3

	Part-B	**M**	**BTL**	**CO**
11	What are the consequences and effects of corrosion, explain with relevant examples.	7	2	2
12	What is the significance of creating a corrosion map of India, and how can it help in addressing corrosion issues more effectively?	7	3	3
13	What are the necessary steps to be taken for the corrosion test program? Explain	7	2	1
14	Two metals A and B undergo oxidation corrosion in the absence of moisture. A *[illegible]*odes faster and corrosion of B takes place negligibly. Predict their Pilling-Bedworth ratio and give one example for each. Also, state the Pilling-Bedworth rule.	7	3	4
15	Differentiate between the Galvanic series and the electrochemical series, highlighting the key distinctions in their representations and significance	7	*[illegible]*	*[illegible]*
16	Explain the following types of corrosion i) Crevice corrosion ii) Erosion corrosion	7	*[illegible]*	*[illegible]*
17	Enumerate four benefits and constraints associated with corrosion testing techniques.	8	*[illegible]*	*[illegible]*

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars	CO1	CO2	CO3	CO4	L1	L2	*[illegible]* (cut off in the scan)
	Max Marks	16	13	24	07	4	33	

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Academic year 2024-2025 (Odd Sem)

DEPARTMENT OF
COMPUTER SCIENCE AND ENGINEERING

Date	**Feb 2025**	**Maximum Marks**	**50+10**
Course Code	**21CS75IE**	**Duration**	**120 Min**
Sem	**07**	**Improvement CIE**	
PROMPT ENGINEERING (INSTITUTIONAL ELECTVES – II (Group G))			

Q. No.	Questions	M	BT	CO
	QUIZ			
1.	1.1 Role-playing prompts often begin with instructions to the model to act as a specific _____ or character.	01	1	1
	1.2 Which setting adjusts the probability distribution to control randomness in the generated text?	01	2	2
	1.3 Why is providing context in a prompt important, and how does it help the model?	01	2	2
	1.4 What is the significance of providing examples in prompts for complex tasks?	01	1	1
	1.5 What is one method to prevent prompt injection in AI models?	01	1	1
	1.6 How does prompt leaking impact AI security?	01	1	1
	1.7 Name one strategy to prevent AI jailbreaking.	01	1	2
	1.8 Why is human feedback important in reinforcement learning?	01	2	3
	1.9 What role does reward function play in reinforcement learning?	01	1	1
	1.10 Name one key algorithm used in reinforcement learning for planning.	01	1	1
	TEST			
2.	(a)Describe an example where PAL can enhance the performance of a language model.	05	2	3
	(b)Explain the concept of chain-of-thought (CoT) prompting and its significance in reasoning tasks.	05	1	2
3	Classify and explain the types of Large Language Model (LLM) prompts based on the type of information being prompted. Provide examples for each type.	10	1	2
4	What is multimodal prompting in AI models? Discuss its significance and challenges in integrating different types of data (text, images, audio, etc.).	10	1	2
5	Define graph prompting and discuss its role in enhancing reasoning capabilities *[illegible]* language models.	*[illegible]*	*[illegible]*	*[illegible]*
6	How can adversarial attacks such as prompt injection and jailbreaking impact *[illegible]* security of AI systems? Suggest possible mitigation strategies.	*[illegible]*	*[illegible]*	*[illegible]*

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BT-Blooms Taxonomy, CO-Course Outcomes

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	*[illegible]* (cut off)
	Test & Quiz	Max Marks	06	28	16	00	10	32	18	00	

Course Outcomes:

Academic year 2024-2025 (Odd Sem)

CO1:	Demonstrate an understanding of prompt engineering principles including how *[illegible]* structure and phrasing impact the performance of AI models.
CO2:	Design and implement effective prompts- to create and apply prompts for various *[illegible]* language processing (NLP) tasks, such as text generation, summarization, and *[illegible]* AI models.
CO3:	Critically evaluate the effectiveness of prompts - assess the quality and performance *[illegible]* in terms of accuracy, coherence, and relevance, identifying areas for improvement.
CO4:	Apply prompt engineering techniques in real-world scenarios - use prompt engineering *[illegible]* strategies to address practical problems in domains such as education, healthcare, and *[illegible]* demonstrating the applicability of AI-driven solutions.
CO5:	Collaborate on projects involving prompt engineering - work effectively in teams to *[illegible]* implement, and evaluate prompt-based solutions, showcasing their ability to contribute *[illegible]* complex AI-related projects.

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Department of Civil Engineering

Date	19-02-2025	Maximum Marks	50+10
Course Code	**21CV75IF**	Duration	120 Min
Sem	VII	IMPROVEMENT CIE	
INTEGRATED HEALTH MONITORING OF STRUCTURES			

Note: Answer all questions.

QUIZ

Sl. No.	Questions	M	BT	CO
1	Sensors measure physical parameters such as _____, _____, _____ during testing. (any 2)	2	1	2
2	Two main types of Dynamic Analysis are _____ and _____	2	1	4
3	List any two characteristics of Static Response Measurement	2	1	3
4	Mention any two advantages of Dynamic Response methods.	2	1	4
5	List the different NDT methods (any 4)	2	1	4

TEST

Sl. No.	Questions	M	BT	CO
1	Explain the objectives of Static response systems and Discuss the measurement techniques used for static response systems..	10	2	3
2	List and discuss the different types of Dynamic Response Methods	10	2	3
3	Discuss different remote SHM techniques, systems and challenges	10	2	4
4	Describe remote data acquisition systems and discuss the components used	10	3	4
5	Explain different Simulation methods and loading methods of Static Field Testing.	10	3	3

Course outcomes:

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	-	2	32	26	10	30	20			

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Department of Electrical & Electronics Engineering

Academic Year 2024-25 (ODD SEM)

Date	19th Feb 2024	Maximum Marks	50
Course Code	21EE75IH	Duration	90 Mins
Sem	VII Semester	TEST-III	
E-Mobility			

Q.No	Part B – Test Questions	Marks	COs	BT
1a.	Draw and explain the functions of components involved in Low power charger	06	4	2
1b.	Mention voltage and current levels of AC and DC charging.	04	2	1
2.	With the neat sketch of circuit diagram and waveforms explain the operation of Boost Converter for Power Factor Correction.	~~06~~ 10	4	3
3a.	Write the various EV charging standards and discuss briefly about ***SAE J1772.***	05	2	2
3b.	Discuss briefly about wireless charging of electric vehicle.	05	2	2
4.	Discuss the following with respect considerations for sizing of drive system. (a) Sizing the propulsion motor (b) Matching the electric machine and the internal combustion engine (ICE).	10	4	2
5a.	Discuss briefly about typical CAN layout in electric vehicle.	05	3	3
5b.	Explain about CAN message frame.	05	3	3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	-	14	10	26	4	26	16	-	-	-

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(The struck-out mark for Q2 was corrected by hand on the original.)

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Department of Electrical & Electronics Engineering

Academic Year 2024-25 (ODD SEM)

Date	19th Feb 2024	Maximum Marks	*[illegible]*
Course Code	21EE75IH	Duration	*[illegible]*
Sem	VII Semester	QUIZ-III	

E-Mobility

Q.No.	Part A – Quiz Questions	Marks	COs	BT
1.	What are the key requirements of an EV charging system?	02	2	*[illegible]*
2.	Why is power factor correction important in battery charging?	02	2	*[illegible]*
3.	What is the function of an AC-DC converter in an EV charger?	02	2	*[illegible]*
4.	What is the role of a communication protocol in EV charging?	02	3	*[illegible]*
5.	What role does an onboard charger play in an EV?	02	3	*[illegible]*

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	-	06	04	-	06	03	01	-	*[illegible]*	*[illegible]*

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Department of Electronics & Telecommunication Engineering

Date	19th FEB. 2025	Maximum Marks/Quiz	50+10
Course Code/Title	21ET75IK	Duration	120 Min
Sem	VII Semester	CIE-III	

SPACE TECHNOLOGY AND APPLICATIONS

Q.No	PART A	M	BT	CO
1.	Define Multispectral Image.	1	L1	CO3
2.	What is Spatial and Spectral resolution?	1	L2	CO3
3.	Mention few applications of Remote sensing	1	L1	CO4
4.	Who was the first person to walk on the moon?	1	L1	CO4
5.	Define Space debris.	1	L1	CO4
6.	What is the difference between space shuttle and spacecraft?	1	L2	CO4
7.	Differentiate between the US Apollo Mission and the India Chandrayaan -1 mission?	2	L2	CO4
8.	What are the most common applications of satellite in day to day life? Mention any four.	2	L2	CO1
	PART B			
1.	Explain how image processing and analysis applied in remote sensing. Illustrate relevant techniques used.	10	L2	CO3
2.	Explain different types of remote sensing cameras by mentioning their various characteristics and differences.	10	L2	CO3
3.	Describe at least two remote sensing satellites launched by India and highlight their applications.	10	L2	CO4
4.a)	What are the Procedures adopted for Disaster and flood management system and explain with the help of a neat diagram.	05	L2	C3
b)	Write a short note on “Inter space communication system”	05	L2	C3
5.	Give explanatory notes on: i) Planetary Payloads ii) Zero gravity experiments.	10	L2	CO4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	01	—	32	17	06	44	–	–	–	–

Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**19 FEB 2025**	**Maximum Marks**	**65**
Course Code	**21IM75IM**	**Duration**	**120 Min**
Sem	**VII**	**Improvement Test**	
PROJECT MANAGEMENT			

Note:

1. Answer all the Questions.

PART – A

Sl. No.	**Questions**	**M**	**BT**	**CO**
1	Which costs are included in project life cycle costing?	02	L2	CO1
2	What are the four main stages of Product Lifecycle Management (PLM)?	02	L2	CO1
3	What is resource levelling in project scheduling?	02	L2	CO2
4	What are the two major network-based scheduling techniques?	02	L2	CO1
5	What is project implementation scheduling?	02	L2	CO2

PART – B

Sl. No.	**Questions**	**M**	**BT**	**CO**
1.	A manufacturing company is launching a new product and needs to plan its project schedule efficiently. As a project manager, explain the steps involved in project implementation scheduling and discuss how effective time management and resource allocation methods can optimize the project timeline.	10	L3	CO3
2.	A construction company is implementing a new infrastructure project. How can different scheduling techniques and PLM (Product Lifecycle Management) concepts be used to optimize project planning? Additionally, explain the significance of life cycle costing in project management.	10	L3	CO3
3.	A project consists of the following activities with their estimated durations (in days) and dependencies: **Activity** / **Predecessor(s)** / **Duration (Days)** A / - / 5 B / A / 4 C / A / 6 D / B / 3 E / B, C / 7 F / D, E. / 5 **Tasks:** 1. Construct the Project Network Diagram. 2. Identify the Earliest Start Time (ES), Earliest Finish Time (EF), Latest Start Time (LS), and Latest Finish Time (LF) for each activity. 3. Determine the critical path and the total project duration.	15	L4	CO4

Academic year 2024-2025 (ODD Sem)

4.	A project consists of the following activities with their estimated time durations in weeks: **Tasks:** 1. Construct the Project Network Diagram. 2. Compute the Expected Time (TE) for each activity 3. Identify the Critical Path and the Project Completion Time.	*[illegible]*	*[illegible]*	*[illegible]*

Activity	Predecessor	Optimistic (a)	Most Likely (m)	Pessimistic (b)
A	-	2	4	6
B	A	3	6	9
C	A	4	5	6
D	B	2	3	8
E	C	3	4	5
F	D,E	4	6	8

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	
	Max Marks	Quiz	6	4					10		*[illegible]* (cut off)
		Test			20	30				20	

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Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
MECHANICAL ENGINEERING

Date	**19/02/2025**	**Maximum Marks**	**10 (Q) + 50 (T)**
Course Code	**21ME75IO**	**Duration**	**120 Min.**
Sem	**VII**	**CIE-III (Improvement Quiz and Test)**	
NUCLEAR ENGINEERING			

Answer all the Questions.

#	Questions	M	BT	CO
	Part – A (Quiz)			
1	A nuclear unit becoming critical means ……	01	L1	01
2	Moderator in nuclear plants is used to ……	01	L1	01
3	The most commonly used moderator in nuclear plants is ……	01	L1	01
4	Breeder reactor has a conversion ratio of ……	01	L2	01
5	One gram of uranium will produce energy equivalent to approximately ……tonne of high-grade coal.	01	L2	01
6	The commonly used material for shielding is ……	01	L1	01
7	The main interest of shielding in nuclear reactor is protection against ……	01	L1	01
8	Reflector in nuclear plants is used to ……	01	L1	01
9	The energy required to be applied to a radioactive nucleus for the emission of a neutron is….	01	L2	01
10	Enriched uranium is one in which ……	01	L2	01
	Part – B			
1	Explain with a sketch the elements of a nuclear reactor.	10	L3	02
2	Explain the factors that control the site selection of nuclear power plant.	10	L4	02
3	Give the classification of nuclear reactors using various parameters.	10	L5	03
4	A 230 g piece of boron (mol wt. 10) absorbs thermal neutrons at the rate of 9.57 x 10^{13} per (cm^3 s). Boron density is 2.3 g/cm^3. Find (a) the thermal neutron flux, and (b) the average distance that a neutron travels before it is absorbed. For thermal neutrons, σ_a = 755 barns and σ_s = 4.0 barns.	10	L5	03
5	A reactor is fuelled with 100 tonnes of natural uranium (atomic mass 238.05) in which the average thermal neutron (2200 m/s) flux is 10^{13} neutrons/cm^2s, The 2200 m/s cross section of U-235 (atomic mass 235.04) are: σ_f = 579 barns and σ_c = 101 barns. The energy release per fission is 200 MeV and 0.715% of natural uranium is U-235. Calculate (a) the rating of the reactor in MW/tonne, (b) the rate of consumption of U-235 per day.	10	L6	04

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	MaxMarks	10	20	20	10	06	04	10	10	20	10

*****All The Best*****

Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
Humanities and Social Sciences

Date	**19th February 2025**	**Maximum Marks**	**10 + 50**
Course Code	**21HS75IQ**	**Duration**	**30 + 90 Min**
Sem	**VII**	**CIE – III- Improvement Test**	
Cognitive Psychology			

Note: Answer all the Questions.

Sl. No.	**Part – A- Quiz Questions**	**10 Marks M**	**BT**	**CO**
1.1	What properties characterize language?	2	L1	CO1
1.2	What is the difference between phonemes and morphemes?	2	L1	CO1
1.3	Which processes can be impaired in dyslexia?	2	L1	CO2
1.4	What are the fundamental structures and processes within the brain?	2	L2	CO2
1.5	Why is the study of brain disorders useful for cognitive psychologists?	2	L2	CO2
	Part – B	**50 Marks**		
Q. No	Questions	**M**	**BT**	**CO**
2	Compare and contrast the kinds of understandings that can be gained by studying speech errors made by healthy people with those that can be gained by studying the language produced by people who have particular brain lesions.	10	L1	CO1
3	Explain in detail how cognitive psychologists should interpret evidence of linguistic universals when considering the linguistic-relativity hypothesis.	10	L2	CO2
4	Researchers already are aware that a deficit of a neurotransmitter, acetylcholine, in the hippocampus is linked to Alzheimer's disease. Given the difficulty of reaching the hippocampus without causing other kinds of brain damage, how might researchers try to treat Alzheimer's disease?	10	L3	CO2
5	What is an area of cognition that could be studied effectively by viewing the structure or function of the human brain? Describe how a researcher might use one of the techniques to study that area of cognition.	10	L1	CO1
6	Explain some of the reasons that researchers are interested in finding out the localization of function in the human brain.	10	L2	CO2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	10	20	20	–	20	20	10	–	–	–

Academic year 2024-2025 (ODD SEM)

DEPARTMENT OF

INDUSTRIAL ENGINEERING& MANAGEMENT

Date	**19/02/2025**	**Maximum Marks**	**10+50**
Course Code	**21HS75IR**	**Duration**	**120 Mins**
Sem	**VII**	**IMPROVEMENT TEST**	
PRINCIPLES & PRACTICES OF CYBER LAW			

Note: Answer all the Questions.

Q.No.	**PART- A**	M	BT	CO
1	Name the two different types of data with their meaning	2	3	CO2
2	Define Meta-Data.	2	2	CO3
3	What are Forensic Tools ?	2	2	CO2
4	What is an FIR in the context of Cyber Crime Investigation?	2	2	CO2
5	What is a Charge Sheet in Cyber Crime Investigation?	2	2	CO2
	PART -B			
1	Define Cyber Crime Cells (CCCs) and discuss their role in the investigation of cybercrimes.	10	02	CO3
2	Explain the structure and key functions of Cyber Crime Cells in India.	10	02	CO3
3	Explain the process of cyber crime investigation in detail, including the role of law enforcement agencies, evidence collection, and legal procedures.	10	03	CO3
4	Discuss the need to protect data in cyberspace and the types of data that require protection.	10	03	CO2
5a	Discuss the General Data Protection Regulations (GDPR) 2016.	05	03	CO2
5b	How can organizations ensure compliance with data protection regulations?	05	03	CO2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	-	28	32	-	-	18	42	-	-	-

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